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Review article

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Mechanical properties of human tumour tissues and their implications for cancer development

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Supplementary Information for

Mechanical properties of human tumour tissues and their implications for cancer development

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Supplementary Tables

Supplementary Table 1. Mechanical properties of distinct human cancer cell types reported in the literature.

Cancer	Method	Properties (Type)	Normal	Malignant	Reference
Bladder	AFM: Adhesion experiments	Adhesion force (F_{adh})	HCV29 N-cadherin unbinding force: F_{adh} = 26.1±7.1 pN	T24 N-cadherin unbinding force: F_{adh} = 61.7±14.6 pN	[95]
Bladder	AFM: Colloidal probe	Young's modulus (E)	RT4: E ~ 500 Pa (median)	RT112: E ~ 400 Pa (median) T24: E ~ 360 Pa (median)	[193]
Bladder	AFM: Conical probe	Young's modulus (E)	HCV29 cells (bimodal): E =16±0.9 kPa and E = 33±2 kPa	HTB-9 cells (bimodal): E = 3±0.1 kPa and E = 6.4±0.5 kPa HT-1376 cells: E = 5.2±0.1 kPa T24 cells (bimodal): E = 2.9±0.5 kPa and E = 6.2±0.6 kPa	[164]
Bladder	AFM: Conical probe	Young's modulus (E)	Hu609: E = 12.9±4.8 kPa HCV29: E = 10±4.6 kPa	BC3726: E = 1.4±0.7 kPa T24: E = 1±0.5 kPa HU456: E = 0.4±0.3 kPa	[191]
Bladder	AFM: Conical probe	Young's modulus (E)	HCV29: E = 3.09±0.78 kPa	T24: E = 0.78±0.2 kPa	[95]
Bladder	AFM: Paraboloid probe	Young's modulus (E)	Hu609: E = 12.88±4.83 kPa	T24: E = 0.99±0.47 kPa	[194]
Bladder	AFM: Paraboloid probe	Young's modulus (E)	HCV29: E = 10.3±3.4 kPa	T24: E = 1.8±0.6 kPa	[195]
Bladder	AFM: Pyramidal probe	Young's modulus (E) Viscoelastic loss factor (λ)	HCV29: Single cells: E = 6.8±0.5 kPa	T24: Single cells: E = 2.7±0.2 kPa	[76]

			Monolayer: E = 16.5±1 kPa Overall loss factor: λ = 0.21±0.05	Monolayer: E = 13±0.9 kPa T24 overall loss factor: λ = 0.28±0.07 HT1376 Single cells: E = 5.3±0.4 kPa Monolayer: E = 5.3±0.3 kPa HT1376 overall loss factor: λ = 0.38±0.16	
Bladder	AFM: Pyramidal probe	Young's modulus (E) Adhesion energy (A)	SV-HUC-1: E = 23.7±1.2 kPa A = 4.04±0.71*10^{-16} J and A = 9.91±0.87*10^{-16} J (bimodal)	MGH-U1: E = 1.68±0.42 kPa and E = 3.48±4.31 kPa (bimodal) A = 26.34±0.61*10^{-16} J and A = 44.59±0.94*10^{-16} J (bimodal)	[57]
Bladder	Optical tweezers microrheology	Shear modulus (G)	Normal human urothelial (NHU) cells: G ~ 50 Pa	RT112: G ~ 35 Pa KU19-19: G ~ 15 Pa	[209]
Breast	AFM: Adhesion experiments	Adhesion force (F_{adh}) Binding probability	MCF-10A: F_{adh} = 141±39 pN binding probability = 5.4±1.8%	MCF-10AT (pre-malignant): F_{adh} = 139±64 pN; binding probability = 11.6±1.9% MCF-10A1a (malignant): F_{adh} = 129±38 pN binding probability = 27.5±5.2%	[94]
Breast	AFM: Conical probe	Young's modulus (E)	184A: E = 2.26±0.56 kPa	T47D: E = 1.2±0.28 kPa MCF-7:	[24]

				E = 1.24±0.46 kPa	
Breast	AFM: Conical probe	Young's modulus (E)	HBL-100: E = 95.4±43.6 kPa	MCF-7: E = 87.3±47.8 kPa MDA-MD-231: E = 55.6±20.1 kPa	[178]
Breast	AFM: Conical probe	Young's modulus (E)	Normal cells: E = 1.93±0.5 kPa	Cancer cells: E = 0.5±0.08 kPa	[177]
Breast	AFM: Paraboloid probe	Young's modulus (E)	MCF10A: (plasma membrane); E = 5.3±1.1 kPa (cortex); E = 37.5±18.2 kPa	MCF10AT (pre-malignant): (plasma membrane); E = 3.7±1.3 kPa (cortex); E = 20.5±5 kPa MCF10CA1a (malignant): (plasma membrane); E = 10.7±3.1 kPa (cortex); E = 15.9±2.7 kPa	[94]
Breast	AFM: Paraboloid probe	Young's modulus (E)	MCF-10A: (cytocortex); E ~ 40 kPa (plasma membrane); E ~ 5 kPa	MCF-10AT (pre-malignant): (cytocortex); E ~ 20 kPa (plasma membrane); E ~ 3 kPa MCF10CA1a (malignant): (cytocortex); E ~ 16 kPa (plasma membrane); E ~ 40 kPa	[179]
Breast	AFM: Pyramidal probe	Young's modulus (E) Viscosity (η)	MCF-10A: E = 1231.07 ± 764.65 Pa η = 22.37±9.62 Pa s	MCF-7: E = 487.44 ±172.67 Pa η =17.31±5.78 Pa s	[160]
Breast [a]	AFM: Pyramidal probe	Viscosity (η)	MCF-10A: η = 5.3±0.1 Pa s	MCF-7: η = 3.44±0.1 Pa s	[210]

				MDA-MD-231: η = 8.69±0.21 Pa s	
Breast	AFM: Pyramidal Probe	Young's modulus (E)	MCF-10A: E = 0.7±0.3 kPa (1 Hz); E = 250±100 kPa (PFM)	MCF-7: E = 0.5±0.1 kPa (1 Hz); E = 28±12 kPa (PFM) MDA-MD-231: E = 0.3±0.1 kPa (1 Hz); E = 25±13 kPa (PFM)	[33]
Breast	AFM: spherical indenter	Elastic modulus (E) Creep time constant (τ) Viscosity (η)	MCF-10A: E = 341.91 ± 97.98 Pa η = 10.2±3.6 mPa s τ = 0.861 ± 0.177 s	MDA-MD-231: E = 277.32 ± 63.13 Pa η = 7.4±2.2 mPa s τ = 0.773±0.151 s MCF-7: E= 285.13±127.03 Pa η = 4.8±2.1 mPa s τ = 0.484±0.002 s	[180]
Breast	AFM: Spherical probe	Viscosity (η)	MCF-10A: η ~ 60 Pa s	MDA-MD-231: η ~ 30 Pa s	[181]
Breast	AFM: Spherical probe	Young's modulus (E)	MCF-10A Cytoskeleton: E = 690.85±432.85 Pa MCF-10A Nucleus: E = 594.66±540.46 Pa	MCF-7 Cytoskeleton: E = 394.89±295.97 Pa MCF-7 Nucleus: E = 399.01±117.16 Pa MDA-MD-231 Cytoskeleton: E = 103.42±89.45 Pa MDA-MD-231 Nucleus: E = 157.7±78.55 Pa	[182]
Breast	AFM: Spherical probe	Young's modulus (E, 1Hz indentation)	MCF-10A: E ~ 1100 Pa	MCF-7: E ~ 550 Pa	[159]
Breast	AFM: Spherical probe	Young's modulus (E)	MCF-10A (RPMI + 10%FBS): E = 1.11±0.85 kPa	MDA-MB-231 (RPMI+10%FBS): E = 0.5±0.35 kPa	[183]

Breast	AFM: Spherical probe	Young's modulus (E)	MCF-10A: E = 1.13±0.84 kPa	MDA-MB-231: E = 0.51±0.35 kPa	[184]
Breast	AFM: Tipless probes (nonadherent) and Spherical probes (adherent)	Young's modulus (E)	Spherical MCF-10A: E = 107.9±159.68 Pa Adherent MCF-10A: E = 165.92±70.26 Pa	Spherical MCF-7: E = 117.48±134.17 Pa Spherical MDA-MB-231: E = 91.34±82.16 Pa Adherent MCF-7: E = 129.28±52.65 Pa Adherent MDA-MB-231: E = 132±52.22 Pa	[185]
Breast	MEMS Resonant Sensor	Elastic modulus (E) Viscosity (η)	MCF-10A: E = 398±104.1 Pa η =10.2±2.7 mPa s	MCF-7: E = 275.2±157.4 Pa η = 7.4±4.2 mPa s MDA-MD-231: E = 257.5±98.4 Pa η =8.1±3.1 mPa s	[180]
Breast	Microfluidic assay	Young's modulus (E)	MCF-10A: E = 441.4±65.3 Pa	MDA-MB-231: E = 206.2 ± 23.1 Pa	[186]
Breast [a]	Microfluidic assay	Entry time (T_E) Elongation Index (EI)	MCF-10A: T_E = 1.698±0.201 s EI = 1.231±0.012	MCF-7: T_E = 0.433±0.045 s EI = 1.281±0.015	[211]
Breast	Optical Tweezer microscopy	Young's modulus (E)	HBL-100: E = 23.5±10.6 kPa	MCF-7: E = 30.2±15 kPa MDA-MD-231: E = 12.6±6.1 kPa	[178]
Breast	Optical tweezers	Deformability (D)	MCF-10: D = 10.5±0.8%	MDA-MB-231: D = 33.7±1.4%	[212]
Breast	Optical tweezers	Tether force (F_T)	MCF-10A: F_T ~ 32 pN	MCF7: F_T ~ 28 pN MDA-MD-231: (ruffling); F_T ~ 22 pN (blebbling);	[85]

				F_T ~ 25 pN	
Breast	Optical tweezers-based microrheology	Shear modulus (G_o) Storage modulus (G') Loss modulus (G'') Rigidity Index (RI) Spring constant (k) Viscosity (η)	MCF-10A: G_o ~ 3.8 Pa G' ~ 3.6 Pa G'' ~ 1.6 Pa RI ~ 0.45 k ~ 160 pN/μm η ~ 18 Pa s	MDA-MB-231 G_o ~ 1.8 Pa G' ~ 1.8 Pa G'' ~ 0.1 Pa RI ~ 0.25 k ~ 50 pN/μm η ~ 13 Pa s	[77]
Breast	Traction force microscopy	Traction force (F_T)	M10: F_T ~ 200 nN	MCF-7: F_T ~ 100 nN	[213]
Breast	Traction force microscopy	Traction force (F_T)	MCF-10A: F_T ~ 6 nN	MCF-7: F_T ~ 14 nN	[214]
Breast (patient-derived cells)	AFM: Conical probe	Young's modulus (E)	Normal breast: (Case 10) E = 1.7±0.95 kPa	Metastatic breast cancer: (Case 10) E = 0.2±0.1 kPa	[161]
Cervical	AFM: Conical probe	Young's modulus (E)	END1/E6E7 Central: E = 5.5±0.54 kPa Peripheral: E = 4.11±1.09 kPa	HeLa Ccentral: E = 2.48±0.5 kPa Peripheral: E = 2.44±0.7 kPa	[215]
Cervical	AFM: Conical probe	Young's modulus (E)	CRL2614: (small indentation); E = 1.26±0.062 kPa (moderate indentation); E = 2.49±0.091 kPa (large indentation); E = 3.98±0.09 kPa	CaSki: (small indentation); E = 0.41±0.065 kPa (moderate indentation); E = 1.83±0.08 kPa (large indentation); E = 3.86±0.28 kPa	[216]
Cervical	AFM: Spherical probe	Young's modulus (E)	Normal cells: E = 1.6±0.6 kPa	Cancer cells: E = 1.4±0.48 kPa	[52]
Cervical (patient-derived)	AFM: Adhesion experiment	Adhesion force (F_{adh})	Normal cells adhesion with silica probe: F_{adh} = 0.39±0.2 nN	Cancer cells adhesion with silica probe: F_{adh} = 0.54±0.16 nN	[217]
Cervical (patient-derived)	AFM: Spherical probe	Young's modulus (E)	Normal cells: E = 2.05±0.48 kPa	Cancer cells: E = 2.8±1.7 kPa	[218]

Esophageal	AFM: Spherical probe	Young's modulus (E)	EPC2: E = 4.72 kPa (modal)	CP-A: E = 3.08 kPa (modal) CP-D: E = 2.64 kPa (modal)	[219]
Fibrosarcoma	AFM: Spherical probe	Young's modulus (E)	Human mammary fibroblasts: E ~ 1 kPa	HT1080: E ~ 0.5 kPa	[220]
Kidney	AFM: Pyramidal probe	Young's modulus (E) Viscosity (η)	RC124: (median elasticity): E = 9.38 kPa (median viscosity): η = 69.6 Pa s	ACHN: (median elasticity): E = 2.48 kPa (median viscosity): η = 24.9 Pa s A498: (median elasticity): E = 7.41 kPa (median viscosity): η = 28.1 Pa s	[221]
Kidney	AFM: Pyramidal probe	Young's modulus (E)	RC124 (petri dish): E ~ 2000 Pa RC124 (patterned PDMS): E ~2500 Pa	A498 (petri dish): E ~ 1000 Pa A498(patterned PDMS): E ~ 800 Pa ACHN metastatic (petri dish): E ~ 1800 Pa ACHN metastatic (patterned PDMS): E ~ 1600 Pa	[222]
Liver	Micropipette aspiration	Elastic (k_1 and k_2) and viscous (η) components	Hepatocytes: k_1 = 87.5±12.1 N/m^2 k_2 = 33.3±10.3 N/m^2 η = 5.9±3 Pa s	HCC: k_1 = 103.6±12.6 N/m^2 k_2 = 42.5±10.4 N/m^2 η = 4.5±1.9 Pa s	[78]
Lung	AFM: Conical probe	Young's modulus (E)	Normal lung: (Case 1); E = 2.1±0.79 kPa (Case 2);	Metastatic lung cancer: (Case 1); E = 0.56±0.09 kPa (Case 2);	[177]

			2.05±0.87 kPa	E = 0.52±0.12 kPa	
Lung, patient-derived cells	AFM: Conical probe	Young's modulus (E)	Normal lung: (Case 2); E = 2.8 ± 1.3 kPa (Case 4); E = 1.7±0.5 kPa (Case 9); E = 2.9± 1.1 kPa	Lung cancer: (Case 2); E = 0.5±0.1 kPa (Case 4); E = 0.3±0.1 kPa (Case 9); E = 0.5±0.1 kPa	[161]
Melanoma	AFM: Conical probe	Young's modulus (E)	Healthy melanocytes (HEMa): E = 14.3±0.8 kPa	VGP (Primary tumor): E = 9.91±0.55 kPa Skin metastasis (WM239): E = 7.8±0.61 kPa Lung metastasis (A375P): E = 5.7±0.61 kPa	[196]
Mesothelial	AFM: Conical probe	Young's modulus (E)	Fluorescent confirmed "benign" cells (overall): E = 1.97±0.7 kPa Centrifuged "normal" cells: E = 2.73±1.3 kPa	Fluorescent confirmed "tumor" cells (overall): E = 0.53±0.1 kPa Centrifuged "cancer" cells: E = 0.5±0.26 kPa	[177]
Oral squamous cell	Optical stretcher	Compliance (δ)	OKF-4: $\delta = 0.69 \pm 0.46 * 10^{-2}$ Pa^{-1} OKF-6: $\delta = 0.75 \pm 0.49 * 10^{-2}$ Pa^{-1}	CAL-33: $\delta = 1.99 \pm 1.63 * 10^{-2}$ Pa^{-1} CAL-27: $\delta = 1.59 \pm 0.75 * 10^{-2}$ Pa^{-1} BHY: $\delta = 1.86 \pm 0.76 * 10^{-2}$ Pa^{-1} HN: $\delta = 1.75 \pm 0.76 * 10^{-2}$ Pa^{-1}	[66]
Ovarian	AFM: Spherical probe	Young's modulus (E)	IOSE: E = 2.472±2.048 kPa	OVCAR4: E = 1.12±0.865 kPa HEY: E = 0.884±0.529 kPa OVCAR3: E = 0.576±0.236 kPa	[162]

				HEYA8: E = 0.494±0.222 kPa	
Ovarian, patient-derived cells	AFM: Conical probe	Young's modulus (E)	Normal ovaries: (Case 3); E = 2.8±1.7 kPa (Case 5); E = 2.5±1.1 kPa (Case 6); E = 2.4±1 kPa (Case 7); E = 2.4±1.1 kPa	Ovarian cancer cells: (Case 3); E = 0.5±0.1 kPa (Case 5); E = 0.5±0.1 kPa (Case 6); E = 0.4±0.1 kPa (Case 7); E = 0.4±0.1 kPa	[161]
Pancreatic	AFM: Paraboloid probe	Young's modulus (E)	HPNE: E ~ 40 kPa	AsPC1: E ~ 5 kPa	[15]
Pancreatic [b]	AFM: Pyramidal probe	Young's modulus (E)	HPDE: E = 3.7±1.2 kPa	Panc-1: E = 2.4±1.1 kPa MIA PaCa-2: E = 1.7±1 kPa HS766T: E = 3±2.0 kPa	[35]
Pancreatic, patient-derived cells	AFM: Conical probe	Young's modulus (E)	Normal pancreas: E = 2.7±1.6 kPa	Metastatic PDAC: E = 0.6±0.1 kPa	[161]
Pleural effusion	AFM: Conical probe	Young's modulus (E) Adhesion force (F_{adh})	Benign cells (cytospin): E = 2.53±1.3 kPa Benign cells (ex vivo): E = 1.97±0.7 kPa Normal cell rupture force (overall): F_{adh} = 51.1±15.2 pN Sample Ranges: Normal stiffness: E = 0.54±0.12 kPa to E = 2.1±0.79 kPa F_{adh} = 36±8 pN to F_{adh} = 60.7±19.6 pN	Tumor cells (cytospin): E = 0.38±0.2 kPa Tumor cells (ex vivo): E = 0.53±0.1 kPa Tumor cell rupture force (overall): F_{adh} = 34.2±5.3 pN Sample Ranges: Cancer stiffness: E = 0.5±0.08 kPa to E = 0.56±0.09 kPa F_{adh} = 29.2±1.9 pN to F_{adh} = 37.1±4 pN	[223]
Prostate	AFM: Conical Probe	Young's modulus (E)	PZHPV-7:	LNCaP:	[24]

			E = 3.09±0.84 kPa	E = 0.45±0.21 kPa DU-145: E = 1.36±0.42 kPa PC-3: E = 1.95±0.47 kPa	
Prostate	AFM: Pyramidal probe	Young's modulus (E) Viscosity (η)	HPV-PZ-7: E ~ 600 Pa (3 kPa substrate); E ~ 1200 Pa (19 kPa substrate); E ~ 1900 Pa (35 kPa substrate) η ~ 15 Pa s (3kPa substrate); η ~ 15 Pa s (19 kPa substrate); η ~ 16 Pa s (35 kPa substrate)	PC-3: E ~ 500 Pa (3 kPa substrate); E ~ 600Pa (19 kPa substrate); E ~ 900 Pa (35 kPa substrate) η~15 Pa s (3 kPa substrate); η ~ 10 Pa s (19kPa substrate); η ~ 8 Pa s (35 kPa substrate)	[79]
Prostate [b]	AFM: Pyramidal probe	Young's modulus (E)	BPH: E = 2797±491 Pa PNT2: E = 1139±320 Pa	PC-3: E = 1401±162 Pa LNCaP: E = 287±52 Pa	[224]
Prostate	AFM: Spherical probe	Young's modulus (E)	Vero: E = 1300±900 Pa	DU145: E = 600±400 Pa	[163]
Prostate cancer associated fibroblasts	Traction force microscopy	Traction force (F_T)	Normal fibroblast: F_T ~ 2 kPa	Cancer associated fibroblast: F_T ~ 3.5 kPa	[225]
Squamous cell carcinoma	AFM: Spherical probe	Young's modulus (E)	Human mammary fibroblasts: E ~ 1 kPa	SCC38: E ~ 0.75 kPa	[220]
Thyroid	AFM: Pyramidal probe	Young's modulus (E)	S748: E ~ 2211-6879 Pa	S277: E ~ 1189-1365 Pa	[200]
Thyroid [c]	AFM: Pyramidal probe	Young's modulus (E) Viscosity (η)	S747: E = 509 Pa (soft gel); E = 845 Pa (stiff gel);	S277: E = 963 Pa (soft gel); E = 867 Pa (stiff gel);	[80]

	E = 1190 Pa (petri dish); η = 224 Pa s (soft gel); η = 516 Pa s (stiff gel); η = 496 Pa s (petri dish)	E = 721 Pa (petri dish); η = 296 Pa s (soft gel); η = 237 Pa s (stiff gel); η = 344 Pa s (petri dish)

AFM, atomic force microscopy; MEMS, micro-electromechanical systems.; PFM, peak force modulation.

± represents mean ± standard deviation, unless specified otherwise.

[a] Data from these references is expressed as the mean ± standard error of the mean (SEM)

[b] Data from these references is expressed as the mean ± 95% confidence interval

[c] Data from these references is expressed as the median value

Supplementary Table 2. Mechanical properties of distinct human cancer tissues reported in the literature.

Cancer	Method	Property	Normal	Malignant	Reference
Bladder	AFM: Pyramidal probe	Young's modulus (E)	HCV29 Spheroid: E = 14.5±2.2 kPa	T24 Spheroid: E = 2.7±0.3 kPa HT1376 Spheroid: E = 2.4±0.2 kPa	[76]
Bladder, patient derived	Nanoindenter	Young's modulus (E)	Healthy bladder muscle tissue: E = 33 kPa (median)	High grade MICB tissue: E = 2 kPa (median)	[192]
Breast	AFM	Young's modulus (E)	Normal adjacent and noninvasive stroma E ~ 400 Pa	Invasive tumor tissue E > 5 kPa	[41]
Breast	AFM: Conical probe	Young's modulus (E)	Healthy breast: E = 1.16±0.2 kPa	Breast cancer (bimodal): E = 0.47±0.15 kPa and E = 1.54±0.17 kPa	[24]
Breast [a]	AFM: Spherical probe	Young's modulus (E)	Low MD: E ~1.8 kPa (multiple tissues) Low MD: E ~ 2.5 kPa (individual breast sample)	High MD: E ~ 2.5 kPa (multiple tissues) High MD: E ~ 4 kPa (individual breast sample)	[187]
Breast	AFM: Spherical probe	Young's modulus (E)	Normal tissue: E = 1.13 ± 0.78 kPa	Invasive tumor (shows multimodal distribution): E1 = 0.57±0.16 kPa E2 = 1.99±0.73 kPa E3 = 5.75±1.62 kPa	[64]
Breast	Axial shear/strain imaging	Normalized axial shear strain area (NASSA)	Benign breast mass (fibroadenoma): NASSA = 0.72	Malignant breast mass (invasive ductal carcinoma): NASSA = 1.64	[226]
Breast	Brillouin microscopy	Young's modulus (E)	Normal spheroid (day 0): E ~ 600 Pa Normal spheroid (day 5): E ~ 1250 Pa	Cancer spheroid (day 0): E ~ 500 Pa Cancer spheroid (day 5): E ~ 1000 Pa	[188]
Breast	Macro-compressor (1 Hz	Young's modulus (E)	5% Precompression: Normal globular tissue:	5% Precompression:	[189]

	loading frequency)		E = 33±11 kPa 20% Precompression: Normal globular tissue: E = 57±19 kPa	Invasive ductal carcinoma: E = 93±33 kPa 20% Precompression: Invasive ductal carcinoma: E = 490±112 kPa	
Breast	Magnetic resonance elastography	Young's modulus (E)	Adjacent Parenchymal tissue: E = 2.5 kPa Fatty tissue: E = 2 kPa	Benign breast lesion: E = 7 kPa Malignant tumor: E = 15.9 kPa	[190]
Colorectal	AFM: Spherical probe	Young's modulus (E)	Normal tissue: E ~ 0.07 kPa	Tumor epithelium: E ~ 0.15 kPa Tumor stroma: E ~ 0.2 kPa	[227]
Colorectal	AFM: Spherical probe	Young's modulus (E)	Control tissue: E = 0.8±0.4 kPa	Tumor tissue: E = 2.4±1.83 kPa	[228]
Colorectal, peritoneal metastasis	AFM: colloidal probe	Young's modulus (E)	Normal tissue: E ~ 3.5 kPa (median)	Neoplastic tissue: E ~ 7 kPa (median)	[193]
Colorectal, peritoneal metastasis	AFM: Spherical probe	Young's modulus (E)	Healthy patient tissue: E ~ 2.5 kPa (Pt1), E ~ 8 kPa (Pt2), E ~ 12.5 kPa (Pt3), E ~ 10 kPa (Pt4), E ~ 2.5 kPa (Pt5)	Neoplastic patient tissue: E ~ 12 kPa (Pt1), E ~ 25 kPa (Pt2), E ~ 30 kPa (Pt3), E ~ 12 kPa (Pt4), E ~ 4 kPa (Pt5)	[229]
Endometrioid	AFM: Conical probe	Young's modulus (E)	Healthy endometrioid: E = 1.27±0.13 kPa	Endometrioid carcinoma: E = 0.57±0.19 kPa	[24]
Glioma	AFM: Spherical probe	Young's modulus (E)	Gliosis: E ~ 10-180 Pa	Low grade glioma: E ~ 50-1400 Pa Glioblastoma: E ~ 70-13500 Pa	[199]
Liver [b]	Ultrasound elastography	Young's modulus (E)	Normal liver: E = 3.9 kPa (3.6-4.1 kPa)	Fibrotic liver: E = 8.3 kPa (3.7-20 kPa) Cirrhosis: E = 17 kPa (4.8-50 kPa)	[230]

				HCC: E = 42 kPa (4.5-265 kPa) Well-differentiated liver: E = 23 kPa (5-38 kPa) Moderately-differentiated liver: E = 37 kPa (4.5-264 kPa) Poorly differentiated liver: E = 52 kPa (7-265 kPa)	
Liver [b]	Ultrasound elastography	Young's modulus (E)	Normal liver (in vivo): E = 10 kPa (1.2-51 kPa) Normal liver (in vitro): E = 5.2 kPa (0.9-18 kPa)	HCC (in vivo): E = 60 kPa (2.4-281 kPa) HCC (in vitro): E = 7.8 kPa (1.5-36 kPa)	[230]
Pancreatic	Macroindentation device	Young's modulus (E) Viscosity (η)	Normal tissue: E = 1.06 ± 0.41 kPa η = 252 ± 134 kPa s	Tumor tissue E = 5.46 ± 3.18 kPa η = 349 ± 222 kPa s	[81]
Prostate	Macro-compressor (1Hz loading frequency)	Young's modulus (E)	2% Precompression: Normal anterior tissue: E = 62±17 kPa 4% Precompression: Normal anterior tissue: E = 63±18 kPa	2% Precompression: Prostate cancer: E = 100±20 kPa 4% Precompression: Prostate cancer: E = 221±32 kPa	[189]
Prostate	Shear wave elastography	Young's modulus (E)	Benign ROIs: Axial peripheral zone: E = 24 kPa median Transition zone: E = 44 kPa median Sagittal peripheral zone: E = 41 kPa median Transition zone: E = 65 kPa median	Cancer ROIs: Axial peripheral zone: E = 46 kPa median Transition zone: E = 87 kPa median Sagittal peripheral zone: E = 61 kPa median Transition zone: E = 92 kPa median	[231]
Prostate	Unconfined compression	Complex modulus (E*) Viscosity (η)	Normal prostate: E* (0.1 Hz) = 3.8±1.8 kPa E* (150 Hz) = 16±5.7 kPa η = 3.6±1.3 kPa s	Cancerous tissue: E* (0.1 Hz) = 7.8±3.3 kPa E* (150 Hz) = 40.6±15.9 kPa η = 8.7±3.4 kPa s	[232]

Thyroid	Shear wave elastography	Young's modulus (E)	Normal tissue (0% precompression): E = 10.3±3.3 kPa Normal (high precompression): E = 21.1±4.2 kPa	Thyroid cancer (0% precompression): E = 22.2±11.9 kPa Cancer (high precompression): E = 97.6±46.8 kPa	[201]
Thyroid	Shear wave elastography	Elasticity index (EI)	Thyroid gland: EI = 15.9±7.6 kPa	Malignant nodules: EI = 150±95 kPa	[202]
Vulva	AFM: Conical probe	Young's modulus (E)	Healthy vulva: E = 3.07±1.6 kPa	Vulvar carcinoma: E = 1.58±0.85 kPa	[24]

AFM, atomic force microscopy; ± represents mean ± standard deviation, unless otherwise specified.

[a] Data from these references are expressed as mean ± standard error of the mean (SEM).

[b] Data from these references are expressed as the mean, and the range of all data points for each group is included in parentheses.

References for data in the supplementary tables

(NOTE: The numbering of these references corresponds to the main text; only the references listed in these tables were included. For a full list of references, please refer to the main manuscript.)

15. Massey, A.E., et al., *Biophysical changes caused by altered MUC13 expression in pancreatic cancer cells.* Micron, 2020. **130**: p. 102822.
24. Lekka, M., et al., *Cancer cell detection in tissue sections using AFM.* Archives of Biochemistry and Biophysics, 2012. **518**(2): p. 151-156.
33. Calzado-Martín, A., et al., *Effect of Actin Organization on the Stiffness of Living Breast Cancer Cells Revealed by Peak-Force Modulation Atomic Force Microscopy.* ACS Nano, 2016. **10**(3): p. 3365-3374.
34. Swaminathan, V., et al., *Mechanical stiffness grades metastatic potential in patient tumor cells and in cancer cell lines.* Cancer Res, 2011. **71**(15): p. 5075-80.
35. Nguyen, A.V., et al., *Stiffness of pancreatic cancer cells is associated with increased invasive potential.* Integrative Biology, 2016. **8**(12): p. 1232-1245.
41. Acerbi, I., et al., *Human breast cancer invasion and aggression correlates with ECM stiffening and immune cell infiltration.* Integrative biology : quantitative biosciences from nano to macro, 2015. **7**(10): p. 1120-1134.
52. Guz, N., et al., *If cell mechanics can be described by elastic modulus: study of different models and probes used in indentation experiments.* Biophysical journal, 2014. **107**(3): p. 564-575.
57. Canetta, E., et al., *Discrimination of bladder cancer cells from normal urothelial cells with high specificity and sensitivity: Combined application of atomic force microscopy and modulated Raman spectroscopy.* Acta Biomaterialia, 2014. **10**(5): p. 2043-2055.
64. Plodinec, M., et al., *The nanomechanical signature of breast cancer.* Nature Nanotechnology, 2012. **7**(11): p. 757-765.
66. Remmerbach, T.W., et al., *Oral cancer diagnosis by mechanical phenotyping.* Cancer Res, 2009. **69**(5): p. 1728-32.
76. Gnanachandran, K., et al., *Discriminating bladder cancer cells through rheological mechanomarkers at cell and spheroid levels.* J Biomech, 2022. **144**: p. 111346.
77. Mandal, K., et al., *Mapping intracellular mechanics on micropatterned substrates.* Proc Natl Acad Sci U S A, 2016. **113**(46): p. E7159-E7168.
78. Zhang, G., et al., *Mechanical properties of hepatocellular carcinoma cells.* World J Gastroenterol, 2002. **8**(2): p. 243-6.
79. Tang, X., et al., *Effects of substrate stiffness on the viscoelasticity and migration of prostate cancer cells examined by atomic force microscopy.* Beilstein J Nanotechnol, 2022. **13**: p. 560-569.
80. Rianna, C. and M. Radmacher, *Comparison of viscoelastic properties of cancer and normal thyroid cells on different stiffness substrates.* Eur Biophys J, 2017. **46**(4): p. 309-324.
81. Rubiano, A., et al., *Viscoelastic properties of human pancreatic tumors and in vitro constructs to mimic mechanical properties.* Acta Biomaterialia, 2018. **67**: p. 331-340.

85. Tsujita, K., et al., *Homeostatic membrane tension constrains cancer cell dissemination by counteracting BAR protein assembly.* Nat Commun, 2021. **12**(1): p. 5930.
94. Dumitru, A.C., et al., *Label-Free Imaging of Cholesterol Assemblies Reveals Hidden Nanomechanics of Breast Cancer Cells.* Adv Sci (Weinh), 2020. **7**(22): p. 2002643.
95. Lekka, M., et al., *Characterization of N-cadherin unbinding properties in non-malignant (HCV29) and malignant (T24) bladder cells.* J Mol Recognit, 2011. **24**(5): p. 833-42.
159. Li, Q.S., et al., *AFM indentation study of breast cancer cells.* Biochem Biophys Res Commun, 2008. **374**(4): p. 609-13.
160. Wang, Y., et al., *Quantitative analysis of the cell-surface roughness and viscoelasticity for breast cancer cells discrimination using atomic force microscopy.* Scanning, 2016. **38**(6): p. 558-563.
161. Cross, S.E., et al., *Green tea extract selectively targets nanomechanics of live metastatic cancer cells.* Nanotechnology, 2011. **22**(21): p. 215101.
162. Xu, W., et al., *Cell stiffness is a biomarker of the metastatic potential of ovarian cancer cells.* PLoS One, 2012. **7**(10): p. e46609.
163. Efremov, Y.M., et al., *Distinct impact of targeted actin cytoskeleton reorganization on mechanical properties of normal and malignant cells.* Biochim Biophys Acta, 2015. **1853**(11 Pt B): p. 3117-25.
164. Ramos, J.R., et al., *The softening of human bladder cancer cells happens at an early stage of the malignancy process.* Beilstein J Nanotechnol, 2014. **5**: p. 447-57.
177. Cross, S.E., et al., *Nanomechanical analysis of cells from cancer patients.* Nat Nanotechnol, 2007. **2**(12): p. 780-3.
178. Coceano, G., et al., *Investigation into local cell mechanics by atomic force microscopy mapping and optical tweezer vertical indentation.* Nanotechnology, 2016. **27**(6): p. 065102.
179. Maja, M., et al., *Surface cholesterol-enriched domains specifically promote invasion of breast cancer cell lines by controlling invadopodia and extracellular matrix degradation.* Cell Mol Life Sci, 2022. **79**(8): p. 417.
180. Corbin, E.A., et al., *Biophysical properties of human breast cancer cells measured using silicon MEMS resonators and atomic force microscopy.* Lab Chip, 2015. **15**(3): p. 839-47.
181. Nematbakhsh, Y., K.T. Pang, and C.T. Lim, *Correlating the viscoelasticity of breast cancer cells with their malignancy.* Convergent Science Physical Oncology, 2017. **3**(3): p. 034003.
182. Fischer, T., A. Hayn, and C.T. Mierke, *Effect of Nuclear Stiffness on Cell Mechanics and Migration of Human Breast Cancer Cells.* Front Cell Dev Biol, 2020. **8**: p. 393.
183. Nikkhah, M., et al., *Evaluation of the influence of growth medium composition on cell elasticity.* J Biomech, 2011. **44**(4): p. 762-6.
184. Nikkhah, M., et al., *The cytoskeletal organization of breast carcinoma and fibroblast cells inside three dimensional (3-D) isotropic silicon microstructures.* Biomaterials, 2010. **31**(16): p. 4552-61.
185. Fischer, T., et al., *Matrix and cellular mechanical properties are the driving factors for facilitating human cancer cell motility into 3D engineered matrices.* Convergent Science Physical Oncology, 2017. **3**(4).
186. Lee, L.M. and A.P. Liu, *A microfluidic pipette array for mechanophenotyping of cancer cells and mechanical gating of mechanosensitive channels.* Lab Chip, 2015. **15**(1): p. 264-73.

187. Northey, J.J., et al., *Stiff stroma increases breast cancer risk by inducing the oncogene ZNF217.* J Clin Invest, 2020. **130**(11): p. 5721-5737.
188. Zhang, J., et al., *Rapid biomechanical imaging at low irradiation level via dual line-scanning Brillouin microscopy.* Nat Methods, 2023. **20**(5): p. 677-681.
189. Krouskop, T.A., et al., *Elastic moduli of breast and prostate tissues under compression.* Ultrason Imaging, 1998. **20**(4): p. 260-74.
190. Patel, B.K., et al., *Association of breast cancer risk, density, and stiffness: global tissue stiffness on breast MR elastography (MRE).* Breast Cancer Res Treat, 2022. **194**(1): p. 79-89.
191. Lekka, M., et al., *Elasticity of normal and cancerous human bladder cells studied by scanning force microscopy.* Eur Biophys J, 1999. **28**(4): p. 312-6.
192. Martinez-Vidal, L., et al., *Micro-mechanical fingerprints of the rat bladder change in actinic cystitis and tumor presence.* Commun Biol, 2023. **6**(1): p. 217.
193. Holuigue, H., et al., *Force Sensing on Cells and Tissues by Atomic Force Microscopy.* Sensors (Basel), 2022. **22**(6).
194. Lekka, M., et al., *Local elastic properties of cells studied by SFM.* Applied Surface Science, 1999. **141**(3-4): p. 345-349.
195. Lekka, M., et al., *The effect of chitosan on stiffness and glycolytic activity of human bladder cells.* Biochimica Et Biophysica Acta-Molecular Cell Research, 2001. **1540**(2): p. 127-136.
196. Bobrowska, J., et al., *Biophysical and Biochemical Characteristics as Complementary Indicators of Melanoma Progression.* Anal Chem, 2019. **91**(15): p. 9885-9892.
199. Miroshnikova, Y.A., et al., *Tissue mechanics promote IDH1-dependent HIF1alpha-tenascin C feedback to regulate glioblastoma aggression.* Nat Cell Biol, 2016. **18**(12): p. 1336-1345.
200. Prabhune, M., et al., *Comparison of mechanical properties of normal and malignant thyroid cells.* Micron, 2012. **43**(12): p. 1267-72.
201. Lam, A.C., et al., *The influence of precompression on elasticity of thyroid nodules estimated by ultrasound shear wave elastography.* Eur Radiol, 2016. **26**(8): p. 2845-52.
202. Sebag, F., et al., *Shear wave elastography: a new ultrasound imaging mode for the differential diagnosis of benign and malignant thyroid nodules.* J Clin Endocrinol Metab, 2010. **95**(12): p. 5281-8.
209. Mandal, K., et al., *Role of a Kinesin Motor in Cancer Cell Mechanics.* Nano Lett, 2019. **19**(11): p. 7691-7702.
210. Rother, J., et al., *Atomic force microscopy-based microrheology reveals significant differences in the viscoelastic response between malign and benign cell lines.* Open Biol, 2014. **4**(5): p. 140046.
211. Hou, H.W., et al., *Deformability study of breast cancer cells using microfluidics.* Biomed Microdevices, 2009. **11**(3): p. 557-64.
212. Guck, J., et al., *Optical Deformability as an Inherent Cell Marker for Testing Malignant Transformation and Metastatic Competence.* Biophysical Journal, 2005. **88**(5): p. 3689-3698.
213. Lin, H.H., et al., *Mechanical phenotype of cancer cells: cell softening and loss of stiffness sensing.* Oncotarget, 2015. **6**(25): p. 20946-58.
214. Li, Z., et al., *Cellular traction forces: a useful parameter in cancer research.* Nanoscale, 2017. **9**(48): p. 19039-19044.
215. Hayashi, K. and M. Iwata, *Stiffness of cancer cells measured with an AFM indentation method.* J Mech Behav Biomed Mater, 2015. **49**: p. 105-11.

216. Zhao, X., et al., *Discrimination Between Cervical Cancer Cells and Normal Cervical Cells Based on Longitudinal Elasticity Using Atomic Force Microscopy.* Nanoscale Res Lett, 2015. **10**(1): p. 482.
217. Iyer, S., et al., *Towards nonspecific detection of malignant cervical cells with fluorescent silica beads.* Small, 2009. **5**(20): p. 2277-84.
218. Iyer, S., et al., *Atomic force microscopy detects differences in the surface brush of normal and cancerous cells.* Nat Nanotechnol, 2009. **4**(6): p. 389-93.
219. Fuhrmann, A., et al., *AFM stiffness nanotomography of normal, metaplastic and dysplastic human esophageal cells.* Phys Biol, 2011. **8**(1): p. 015007.
220. van Helvert, S. and P. Friedl, *Strain Stiffening of Fibrillar Collagen during Individual and Collective Cell Migration Identified by AFM Nanoindentation.* ACS Appl Mater Interfaces, 2016. **8**(34): p. 21946-55.
221. Rebelo, L.M., et al., *Comparison of the viscoelastic properties of cells from different kidney cancer phenotypes measured with atomic force microscopy.* Nanotechnology, 2013. **24**(5): p. 055102.
222. Rianna, C. and M. Radmacher, *Influence of microenvironment topography and stiffness on the mechanics and motility of normal and cancer renal cells.* Nanoscale, 2017. **9**(31): p. 11222-11230.
223. Cross, S.E., et al., *AFM-based analysis of human metastatic cancer cells.* Nanotechnology, 2008. **19**(38): p. 384003.
224. Faria, E.C., et al., *Measurement of elastic properties of prostate cancer cells using AFM.* Analyst, 2008. **133**(11): p. 1498-500.
225. Erdogan, B., et al., *Cancer-associated fibroblasts promote directional cancer cell migration by aligning fibronectin.* J Cell Biol, 2017. **216**(11): p. 3799-3816.
226. Xu, H., et al., *Axial-Shear Strain Imaging for Differentiating Benign and Malignant Breast Masses.* Ultrasound in Medicine & Biology, 2010. **36**(11): p. 1813-1824.
227. Lopez-Crapez, E., et al., *Mechanical signatures of human colon cancers.* Sci Rep, 2022. **12**(1): p. 12475.
228. Brauchle, E., et al., *Biomechanical and biomolecular characterization of extracellular matrix structures in human colon carcinomas.* Matrix Biol, 2018. **68-69**: p. 180-193.
229. Varinelli, L., et al., *Decellularized extracellular matrix as scaffold for cancer organoid cultures of colorectal peritoneal metastases.* J Mol Cell Biol, 2023. **14**(11).
230. Ling, W., et al., *Effects of vascularity and differentiation of hepatocellular carcinoma on tumor and liver stiffness: in vivo and in vitro studies.* Ultrasound Med Biol, 2014. **40**(4): p. 739-46.
231. Rouviere, O., et al., *Stiffness of benign and malignant prostate tissue measured by shear-wave elastography: a preliminary study.* Eur Radiol, 2017. **27**(5): p. 1858-1866.
232. Hoyt, K., et al., *Tissue elasticity properties as biomarkers for prostate cancer.* Cancer Biomark, 2008. **4**(4-5): p. 213-25.